AIST3010/ESTR3112: Numerical Optimization
Final Review

November 29, 2023

Sample Final: Question 1 (True/False)

- Is this statement True or False?

  If a twice-differentiable $f : \mathbb{R}^d \to \mathbb{R}$ has positive definite Hessian $H_f(\mathbf{x}) \succ \mathbf{0}$ at $\mathbf{x} \in \mathbb{R}^d$, then $\mathbf{x}$ is guaranteed to be a local minimum of f.

Sample Final: Question 1 (True/False)

- Is this statement True or False?

  For every matrix $Q \in \mathbb{R}^{d \times d}$ and vector $\mathbf{b} \in \mathbb{R}^d$, every local maximum of the function $f(\mathbf{x}) = \mathbf{x}^\top Q \mathbf{x} + \mathbf{b}^\top \mathbf{x}$ will be a global maximum.

Sample Final: Question 1 (True/False)

- Is this statement True or False?

  KKT conditions are sufficient for local optimality in every linear programming problem.

Sample Final: Question 1 (True/False)

- Is this statement True or False?

  For every initial point $\mathbf{x}^{(0)}$, Newton's method with unit stepsize will find the global optimum of a quadratic function $f(\mathbf{x}) = \mathbf{x}^\top Q\mathbf{x} + \mathbf{b}^\top \mathbf{x} + c$ with a PSD $Q \succeq \mathbf{0}$ in only on iteration.

Sample Final: Question 1 (True/False)

- Is this statement True or False?

  The gradient descent algorithm is a special case of trust region methods.

Sample Final: Question 2

Question 2 (25 Points): Consider the following optimization problem for positive real numbers $N_1, \ldots, N_d > 0$ and natural logarithm $\log(\cdot)$ with base e:

$$\begin{aligned} \min_{\mathbf{x} \in \mathbb{R}^d} \quad & \sum_{i=1}^{d} \log\left(\frac{N_i}{x_i + N_i}\right) \\ \text{subject to} \quad & \mathbf{x} \geq \mathbf{0}, \\ & \sum_{i=1}^{d} x_i = 1. \end{aligned}$$

1. Is this problem a convex optimization task? If so show it is a convex optimization problem, and if not explain why it is a non-convex problem.

Sample Final: Question 2

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- Write the KKT conditions for this optimization problem. Are the KKT conditions sufficient for a locally optimal solution?

Sample Final: Question 3

Question 3 (20 Points): Consider the following constrained optimization problem for a twice-differentiable **convex** objective function $f : \mathbb{R}^d \to \mathbb{R}$, matrix $A \in \mathbb{R}^{p \times d}$, and vector $\mathbf{b} \in \mathbb{R}^p$:

$$\begin{aligned} \min_{\mathbf{x} \in \mathbb{R}^d} \quad & f(\mathbf{x}) \\ \text{subject to} \quad & A\mathbf{x} = \mathbf{b}. \end{aligned}$$

1. Write the KKT conditions for the above optimization problem. Are the KKT conditions sufficient for a locally optimal solution?

Sample Final: Question 3

Question 3 (20 Points): Consider the following constrained optimization problem for a twice-differentiable **convex** objective function $f : \mathbb{R}^d \to \mathbb{R}$, matrix $A \in \mathbb{R}^{p \times d}$, and vector $\mathbf{b} \in \mathbb{R}^p$:

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2. Formulate an unconstrained optimization problem that is equivalent to this optimization task.

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$$\begin{aligned} \min_{\mathbf{x} \in \mathbb{R}^d} \quad & f(\mathbf{x}) \\ \text{subject to} \quad & A\mathbf{x} = \mathbf{b}. \end{aligned}$$

3. Provide the update equation of gradient descent, which derives $\mathbf{x}^{(k+1)}$ from $\mathbf{x}^{(k)}$, for minimizing the equivalent unconstrained problem in Part 2.

Sample Final: Question 3

Question 3 (20 Points): Consider the following constrained optimization problem for a twice-differentiable **convex** objective function $f : \mathbb{R}^d \to \mathbb{R}$, matrix $A \in \mathbb{R}^{p \times d}$, and vector $\mathbf{b} \in \mathbb{R}^p$:

$$\begin{aligned} \min_{\mathbf{x} \in \mathbb{R}^d} \quad & f(\mathbf{x}) \\ \text{subject to} \quad & A\mathbf{x} = \mathbf{b}. \end{aligned}$$

4. Provide the update equation of Newton's method, which derives $\mathbf{x}^{(k+1)}$ from $\mathbf{x}^{(k)}$, for minimizing the equivalent unconstrained problem in Part 2.

Sample Final: Question 3

Question 3 (20 Points): Consider the following constrained optimization problem for a twice-differentiable **convex** objective function $f : \mathbb{R}^d \to \mathbb{R}$, matrix $A \in \mathbb{R}^{p \times d}$, and vector $\mathbf{b} \in \mathbb{R}^p$:

$$\begin{aligned} \min_{\mathbf{x} \in \mathbb{R}^d} \quad & f(\mathbf{x}) \\ \text{subject to} \quad & A\mathbf{x} = \mathbf{b}. \end{aligned}$$

5. Next we want to directly solve the constrained optimization problem using the projected gradient descent method. Write the pseudocode of projected gradient descent with step-size α that is directly applied to the constrained optimization problem.